Nguyen Dinh Truong

AIoT & Robotics Engineer

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Summary

AI Engineering student at VKU with a solid foundation in Machine Learning and Deep Learning. Self-taught in Embedded Systems and Robotics to deploy AI beyond theory — on real hardware. Competed in technical competitions and achieved First Prize at RoboCar 2025 and Second Prize at the Vietnam Electronics Design Competition 2025. Experienced in designing and building complete autonomous systems end-to-end, from embedded firmware and sensor integration to deep learning model deployment.

Education

Artificial Intelligence
Vietnam-Korea University of Information and Communication Technology (VKU)
2023 – 2028

Skills

- **AI / Machine Learning** — CNN, RNN, LSTM, Transformer, Deep Reinforcement Learning, TinyML, Computer Vision, PyTorch
- **Embedded Systems** — ESP32-S3, Raspberry Pi 4, Arduino, PID Control, Encoder, MPU6050, UART, I2C, SPI, C/C++, Python
- **Robotics** — Autonomous Navigation, Lane Detection, Self-balancing Control, Sensor Fusion, OpenCV
- **Wireless & Protocol** — LoRa, UWB RTLS, TCP/IP, MQTT

Projects

RoboCar 2025
Arduino, C++, PID, Servo, IR Sensor

- Designed an autonomous line-following robot using 5 IR sensors to navigate turns, climb stairs, and stop accurately at designated positions
- Integrated HC-SR04 ultrasonic sensor for obstacle detection, automatically reducing speed when approaching stairs and accelerating through
- Built a PS2-controlled robot with a 5-servo arm capable of picking up, carrying, and shooting balls into targets via wireless control
- Programmed automatic speed reduction when entering grab mode for more precise arm operation
- Integrated and coordinated multiple subsystems — DC motors, servo mechanisms, IR and ultrasonic sensors — operating simultaneously under competition time pressure
- 🥇 First Prize — RoboCar 2025 Competition, VKU

AIoT Smart Helmet
ESP32-S3, TinyML, 1D-CNN, UWB, LoRa, Python

- Deployed a 1D-CNN fall detection model directly on ESP32-S3 using TFLite Micro, achieving 96.2% accuracy with sub-30ms latency
- Compressed model from 127KB to 14KB via INT8 post-training quantization, enabling on-device inference within ESP32-S3 internal SRAM
- Integrated multi-sensor fusion: 6-axis IMU (MPU9250), heart rate and SpO2 (MAX30105), body temperature (MAX30205), battery monitoring (INA219), and GPS
- Built a dual-frequency LoRa relay network (uplink f1=433.0MHz, backhaul f2=433.5MHz) extending coverage to 1km, supporting hundreds of nodes without gateway congestion
- Combined UWB TWR positioning (5–30cm accuracy) with LoRa telemetry for continuous real-time worker tracking across indoor and outdoor environments
- Designed local alert logic independent of network connectivity: fall detection or proximity breach under 3m triggers immediate buzzer alert within 30ms
- 🥈 Second Prize — Vietnam Electronics Design Competition 2025

Autonomous Lane-Following Robot — DQN
Raspberry Pi 4, ESP32-S3, PyTorch, OpenCV

- Built a 2-wheel self-balancing robot that learns to follow lanes on real hardware, entirely self-built from scratch
- Designed a 3-tier system: ESP32-S3 runs real-time PID balancing (Encoder + MPU6050), Raspberry Pi 4 handles HSV-based lane detection and Double DQN inference at ~32 FPS, Laptop serves as training server
- Built lane detection pipeline using OpenCV HSV: Gray → OTSU threshold → Morphology → fitLine → extracting offset and angle as DQN input
- Implemented Double DQN with Replay Buffer 20K, Polyak soft update ($\tau=0.005$), and 4-dim state space (offset, angle, prev_offset, prev_angle) — previous values allow the agent to perceive lateral drift velocity
- Applied 2-phase training: Phase 1 bootstraps from human demonstration data, Phase 2 transitions to agent self-learning with epsilon-greedy decay from $\varepsilon=0.3$ to 0.05
- Model weights streamed in real time from training server to Pi via TCP socket throughout the training process

Awards

First Prize
RoboCar 2025 Competition, VKU

Second Prize
Vietnam Electronics Design Competition 2025